

- 4 (a) Define *potential* at a point in an electric field.

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 [1]

- (b) Fig. 4.1. shows the equipotential lines of a non-uniform electric field.

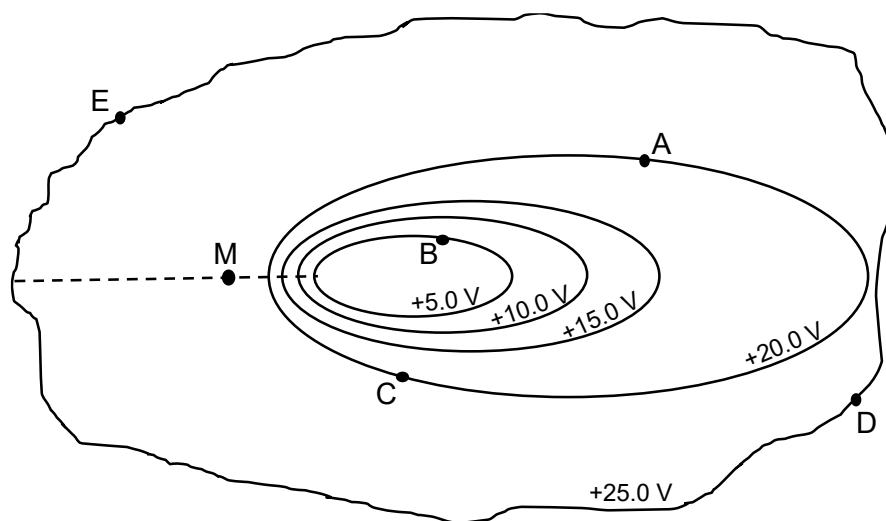


Fig 4.1

- (i) State the potential difference between point A and point B.

potential difference = V [1]

- (ii) Determine the work done to bring a charge of $+5.0 \mu\text{C}$ from point B to point E, via point C and point D.

work done = J [2]

- (iii) Explain whether the work done you have obtained in (ii) is the work done by the electric field or the work done by an external agent.

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 [2]

- (c) (i) State the relation between electric field strength and potential.

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.....[1]

- (ii) Draw an arrow at point **D** in Fig. 4.1 to show the direction of the electric field strength at that point. [1]

- (ii) Another positive charge is released from point M in Fig. 4.1. It moves along the dotted line.

Describe, without further calculation, the motion of this positive charge along the dotted line.

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.....[2]

[Total: 10]